Department of Information Systems and Technologies

CTIS 365: Applied Data Analysis

Semester: SPRING 2025-2026

Lab Guide #8

OBJECTIVES: Correlation, Regression

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Q1. Calculate the **Spearmen correlation** for the following set of scores.

```
x <- c(8, 3, 6, 3, 9, 5, 7)
y <- c(4, 2, 5, 2, 8, 6, 7)
```

Q2. Calculate the **Pearson correlation** by converting the following sample data to z-score.

```
x <- c(12, 7, 15, 10, 9, 14, 6)
y <- c(18, 11, 20, 13, 12, 19, 8)
```

Q3. Use the below shown scores then create the table in the output to find the regression equation between X and Y.

X	Y
5	10
1	4
4	5
7	11
6	15
4	6
3	5
2	0

<u>Example Output:</u>

X	Y	$X - M_X$	$Y - M_Y$	$(X - M_X)^2$	$(Y - M_Y)^2$	$(X - M_X)(Y - M_Y)$
5	10	1	3	1	9	3
1	4	−3	−3	9	9	9
4	5	0	−2	0	4	0
7	11	3	4	9	16	12
6	15	2	8	4	64	16
4	6	0	−1	0	1	0
3	5	−1	−2	1	4	2
2	0	−2	−7	4	49	14
				$SS_X = 28$	$SS_Y = 156$	$SP = 56$